

Gaussian Elimination & Echelon Forms: Practical Reduction Techniques

High-Level Overview

In our previous session, we defined what a **Row Echelon Form (REF)** and a **Reduced Row Echelon Form (RREF)** look like. Now, we put those definitions into practice using **Gaussian Elimination** and **Gauss-Jordan Elimination**.

- **Gaussian Elimination** is the systematic process of applying elementary row operations to reduce any given matrix into **Row Echelon Form (REF)**.
- **Gauss-Jordan Elimination** takes it a step further, pushing the matrix all the way into **Reduced Row Echelon Form (RREF)**.

An essential theorem governing these techniques states: **"Each matrix is row equivalent to one and only one reduced echelon matrix."** No matter what valid sequence of row operations you choose, you will always arrive at the exact same RREF!

Step-by-Step Problem 1: Reducing a 4×4 Matrix to REF

Let's walk through the manual reduction of matrix A into row echelon form.

Given Matrix:

$$A = \begin{bmatrix} 2 & 4 & 3 & -1 \\ 0 & 1 & 8 & 3 \\ 4 & 1 & 3 & 4 \\ 3 & 2 & 1 & 5 \end{bmatrix}$$

Step 1: Create a Leading Pivot of 1 in Row 1

To make calculations cleaner, we scale the first row by dividing by 2:

- Operation: $R_1 \rightarrow \frac{R_1}{2}$

$$\sim \begin{bmatrix} 1 & 2 & 3/2 & -1/2 \\ 0 & 1 & 8 & 3 \\ 4 & 1 & 3 & 4 \\ 3 & 2 & 1 & 5 \end{bmatrix}$$

Step 2: Eliminate Entries Below the First Pivot

We want zeros in the first column under the top-left 1.

- Operations: $R_3 \rightarrow R_3 - 4R_1$ and $R_4 \rightarrow R_4 - 3R_1$

$$\sim \begin{bmatrix} 1 & 2 & 3/2 & -1/2 \\ 0 & 1 & 8 & 3 \\ 0 & -7 & -3 & 6 \\ 0 & -4 & -7/2 & 13/2 \end{bmatrix}$$

Step 3: Clear Column 2 Under the Second Pivot

Our second pivot is already 1 in row 2 (R_{22}). Now, eliminate the entries below it using R_2 :

- Operations: $R_3 \rightarrow R_3 + 7R_2$ and $R_4 \rightarrow R_4 + 4R_2$

$$\sim \begin{bmatrix} 1 & 2 & 3/2 & -1/2 \\ 0 & 1 & 8 & 3 \\ 0 & 0 & 53 & 27 \\ 0 & 0 & 57/2 & 37/2 \end{bmatrix}$$

Step 4: Normalize and Complete REF

Scale row 3 to get a leading entry of 1, then clear the entry below it:

- Operations: $R_3 \rightarrow \frac{R_3}{53}$, followed by $R_4 \rightarrow R_4 - \frac{57}{2}R_3$

$$\sim \begin{bmatrix} 1 & 2 & 3/2 & -1/2 \\ 0 & 1 & 8 & 3 \\ 0 & 0 & 1 & 27/53 \\ 0 & 0 & 0 & 3.98 \end{bmatrix}$$

This final structure satisfies all REF properties: all non-zero rows form a clear staircase pattern with zeros strictly beneath each leading pivot.

Step-by-Step Problem 2: Reducing a Rectangular Matrix

Let's find the row echelon form of a 3×6 matrix B :

$$B = \begin{bmatrix} 4 & 8 & 0 & 0 & -4 & -20 \\ -3 & -5 & 0 & 4 & 6 & 12 \\ 2 & 6 & 0 & 9 & 5 & -14 \end{bmatrix}$$

Step 1: Scale Row 1

Make the leading coefficient 1:

- Operation: $R_1 \rightarrow \frac{R_1}{4}$

$$\sim \begin{bmatrix} 1 & 2 & 0 & 0 & -1 & -5 \\ -3 & -5 & 0 & 4 & 6 & 12 \\ 2 & 6 & 0 & 9 & 5 & -14 \end{bmatrix}$$

Step 2: Clear First Column

Eliminate entries below the first pivot:

- Operations: $R_2 \rightarrow R_2 + 3R_1$ and $R_3 \rightarrow R_3 - 2R_1$

$$\sim \begin{bmatrix} 1 & 2 & 0 & 0 & -1 & -5 \\ 0 & 1 & 0 & 4 & 3 & -3 \\ 0 & 2 & 0 & 9 & 7 & -4 \end{bmatrix}$$

Step 3: Finalize Staircase Form

Clear out the entry under the second pivot:

- Operation: $R_3 \rightarrow R_3 - 2R_2$

$$\sim \begin{bmatrix} 1 & 2 & 0 & 0 & -1 & -5 \\ 0 & 1 & 0 & 4 & 3 & -3 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{bmatrix}$$

This matrix is now successfully in **row-echelon form**.

Column Transformations and CEF

Symmetrically, matrices can be reduced into **Column Echelon Form (CEF)** or **Column Reduced Echelon Form (CREF)** using column operations (denoted by C_i).

For instance, given matrix C , applying targeted column shifts like $C_2 \rightarrow C_2 - 2C_1$, swapping columns, and isolating pivot vectors relocates zero columns to the right and establishes a right-moving column hierarchy:

$$C = \begin{bmatrix} 4 & 0 & 0 & 0 & 0 & 0 \\ -3 & 1 & 0 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Summary Checklist for Rapid Revision

- ☐ Remember that **Gaussian Elimination** stops at Row Echelon Form (REF).
- ☐ Remember that **Gauss-Jordan Elimination** continues until the matrix reaches Reduced Row Echelon Form (RREF).
- ☐ Always prioritize creating leading pivots equal to 1 to minimize fraction arithmetic errors.
- ☐ Verify that every step strictly preserves row equivalence ($\sim$).